

Verifiable Multi-Dimensional Range Queries over Outsourced Datasets in Zero-Knowledge

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Abstract—In data outsourcing applications, a data owner outsources his/her dataset to a remote server who answers queries issued by a client. The server may return incorrect answers to the client, thus a method to verify the correctness of the answers is in need. We focus on verifiable multi-dimensional range queries which are fundamental queries in practice. In [1], Ghosh et al. proposed an efficient method to support verifiable one-dimensional range queries for outsourced datasets of key-value pairs. Their approach is the first to have a zero-knowledge property which means that the client cannot learn anything about the dataset beyond the answers to the queries. However, the approach that directly extends their technique to deal with multi-dimensional range queries is inefficient, requiring $\Omega(2^l)$ storage space of the server, where l is the length of the keys. In this study, we propose an efficient approach for verifiable multi-dimensional range queries in zero-knowledge. Our construction is based on the multi-dimensional hierarchical identity-based encryption (M-HIBE) scheme. We first propose an approach to deal with two-dimensional range queries. Our approach only introduces a multiplicative overhead of l over the storage complexity of [1]. Thus, our approach is efficient. The generalization of the proposed approach to deal with d -dimensional range queries where $d > 2$ is promising.

Index Terms—multi-dimensional range queries, zero-knowledge proof, multi-dimensional hierarchical identity-based encryption, data outsourcing

I. INTRODUCTION

Data outsourcing [2] is an emerging paradigm. It allows data owners to store, manage and disseminate their data in a cost-effective way. In this paradigm, data are outsourced to a third party who is responsible to host the data and provide a variety of functionalities, such as modification and answering queries. Among those functionalities, range queries are fundamental which are motivated by many applications. For example, an internet service provider outsources the storage of network audit logs to an external service provider. An auditor may want to search the logs by querying network flows whose dates and port numbers fall within certain ranges. For instance, the auditor may query for network flows whose dates are within [01/Aug/2019,10/Oct/2019] and the port numbers are within [1, 1023]. Another example would be an individual who stores his/her emails on a remote server and a police may want to query for emails which are sent between certain periods.

Since the answers to range queries are returned by a third party who is not trusted by the data owner, the third party may return incorrect answers. Thus, a mechanism to make the correctness of the answers verifiable is needed. In this study, we focus on supporting multi-dimensional range queries in

a verifiable and privacy-preserving manner. The problem we address is formulated as follows.

In data outsourcing applications, there are typically three parties, i.e. a data owner, a server and a client. The data owner outsources a dataset $\mathcal{D}$ to the server who hosts the outsourced dataset and answers multi-dimensional range queries issued by the client. The dataset $\mathcal{D}$ is modeled as a table containing n tuples and each tuple has d attribute values where $d \geq 1$. Assume each attribute value is an l -bit binary string. $\forall i \in [1, d]$, the i th attribute corresponds to the i th dimension. A d -dimensional range query is expressed as d intervals, i.e. $[l_i, u_i]$ for the i th dimension for i ranging from 1 to d .

The answer to the range query is denoted as $\mathcal{D}'$ which contains tuples whose values on the i th attribute fall within $[l_i, u_i]$ for all $i \in [1, d]$. To support multi-dimensional range queries in a verifiable and privacy-preserving manner, we need to make sure the following two requirements are satisfied.

- 1) The correctness of the answer $\mathcal{D}'$ is verifiable, which requires the server to provide proofs to the client.
- 2) The proofs have zero-knowledge property which means the client learns nothing about $\mathcal{D}$ beyond the answer to the query.

$\mathcal{D}'$ is correct if both of the following conditions hold.

- $\mathcal{D}' \subseteq \mathcal{D}$. This means $\mathcal{D}'$ is a subset of original dataset outsourced by the data owner, thus $\mathcal{D}'$ is not forged.
- No tuple in $\mathcal{D} \cap \bar{\mathcal{D}'}$ satisfies the range query. That means all tuples satisfying the range query are in $\mathcal{D}'$.

To the best of our knowledge, the work in [1] provides an efficient solution for verifiable one-dimensional range queries in zero-knowledge for the first time. Their construction is based on the hierarchical identity-based encryption (HIBE) [3] scheme. However, as the authors point out, the approach which directly extends their technique to multi-dimensional range queries is very inefficient, since it requires $\Omega(2^l)$ secret keys of HIBE to be stored by the server. Thus, designing an efficient method to support verifiable multi-dimensional range queries in zero-knowledge is challenging.

To address the problem, we come up with a solution based on the multi-dimensional hierarchical identity-based encryption (M-HIBE) scheme which extends HIBE [3] to multiple dimensions. We first propose an approach to deal with two-dimensional range queries. The number of secret keys stored by the server is $O(nl^2)$ which is only l times of that in [1]. Therefore, our scheme is efficient.

The rest of this paper is organized as follows. In Section II, we introduce some related works on verifiable multi-dimensional range queries. The system model and security

requirements are described in Section III. In Section IV, technical preliminaries are presented. We introduce our proposed approach in Section V. Finally, the future work is given in Section VI.

II. RELATED WORK

There are many researches conducted to support verifiable multi-dimensional range queries for outsourced data. Unfortunately, they do not have the zero-knowledge property. We list the information leakage of those works as follows.

The leakage of the boundary information is common in previous works. Given a range query, the boundary information is the tuples that are just beyond the queried range. In [4], Martel et al. proposed an approach to support verifiable multi-dimensional range queries by extending one-dimensional Merkle hash trees to multiple dimensions. However, due to the construction of Merkle hash trees, the boundary information is leaked. There are some works addressing verifiable multi-dimensional range queries using signature chaining, such as [5–7]. Those approaches also leak the boundary information. Goodrich et al. constructed a scheme based on signature amortization which unfortunately also leaks the boundary information [8]. Yang et al. proposed a method to deal with verifiable two-dimensional range queries by utilizing an R^* -tree [9] and a structure of hash values similar to Merkle hash trees in [10], where the boundary information is also leaked. In [11], Chen et al. proposed a data structure called local bit matrices that allows a client to verify the correctness of a query result. However, their approach is similar to the signature chaining approach and thus it leaks the boundary information.

In [12], Chen et al. designed a scheme based on canonical range trees which helps to verify the answers to multi-dimensional range queries without leaking boundary information. However, the client learns the number of tuples that are outside of the query range. In [13], Papadopoulos et al. proposed a method for the efficient authentication of d -dimensional range queries via the combination of d separate 1-dimensional proofs and via the set-operation protocol. Their approach lets the server send the partial results which only satisfy the queried range in one dimension to the client. The work in [14] is similar to that of [13], with the main difference being that they additionally support summation queries.

The researches on zero-knowledge sets, e.g. [15–17], provide techniques for zero-knowledge proofs. Their techniques allow the proofs of membership and non-membership of elements in sets to be zero-knowledge. However, it is not straightforward to construct proofs for the answers to multi-dimensional range queries using their techniques.

The approaches proposed for multi-dimensional range queries over encrypted data are promising to provide alternatives to M-HIBE, such as [18–20]. Generally, their techniques allow a server to determine whether encrypted tuples satisfy the queried range. Particularly, in [18], Shi et al. proposed an approach which allows a decryption key corresponding to a range to decrypt any ciphertext intended for a tuple satisfying the range. Their approach works in a similar way of M-HIBE. However, the proofs constructed using their approach instead of M-HIBE cannot guarantee to be zero-knowledge.

III. ADVERSARIAL MODEL AND SECURITY REQUIREMENTS

The adversarial model and the security requirements in this work are the same as that of [1]. We briefly review the adversarial model and security requirements in [1].

A. Adversarial Model

The model consists of three parties, i.e. a *trusted* data owner, an *untrusted* client, and an *untrusted* server. The data owner uploads his/her dataset $\mathcal{D}$ to the server and then goes offline. The server hosts the dataset and answers multi-dimensional range queries issued by the client. The server is untrusted since he/she wants to give incorrect answers without being detected. The client is also untrusted since he/she wants to know more information about the dataset beyond the answers to his/her current and previous range queries. Note that the server and the client are non-colluding. Otherwise, the client can know the whole dataset.

B. Security Requirements

We brief review the security requirements of [1] by describing the following definitions conceptually rather than formally defining them.

- *Completeness.* For any dataset and any multi-dimensional range query, if the data owner and the server executes the protocol honestly, with probability 1, the client will be convinced of the correctness of the answer if the answer is correct.
- *Soundness.* The server can convince the client of an incorrect answer with at most negligible probability.
- *Zero-Knowledge.* The untrusted client cannot learn anything about the dataset beyond what he/she has queried.

IV. PRELIMINARIES AND DEFINITIONS

In this section, we give some preliminaries and definitions that are critical to our construction.

A. Definitions of the canonical coverings of a range

Let $\mathcal{T}_l$ be a full binary tree of with $l + 1$ levels, where the root node (the level 0) is labeled *NULL*. The nodes at the i th level of the tree are i -bit binary strings from 0^i to $2^i - 1$, from left to right. The leaf nodes are l -bit binary strings from 0^l to $2^l - 1$. Fig. 1 shows an example for $\mathcal{T}_3$. A range $[a, b]$ w.r.t. $\mathcal{T}_l$ is the set of contiguous leaf nodes who fall within $[a, b]$ where a, b are l -bit binary strings.

The canonical covering of a range $[a, b]$ is the *minimal* set of nodes, P , of $\mathcal{T}_l$ such that (1) every leaf node in $[a, b]$ is equal to or is the descendant of one of the nodes in P ; (2) for any $z \in P$, in the subtree rooted at z , the leftmost leaf node (denoted as $z(\text{left})$) and the rightmost leaf node (denoted as $z(\text{right})$) satisfy $[z(\text{left}), z(\text{right})] \subseteq [a, b]$. Note that the canonical covering of a range $[a, b]$ is unique. For example, in $\mathcal{T}_3$, the leaf nodes are $\{000, 001, \dots, 111\}$. Given a range $[001, 100]$, the canonical covering of $[001, 100]$ is $[001, 01, 100]$. As shown in Fig. 1, the nodes surrounded by squares are the canonical coverings of the range $[001, 100]$.

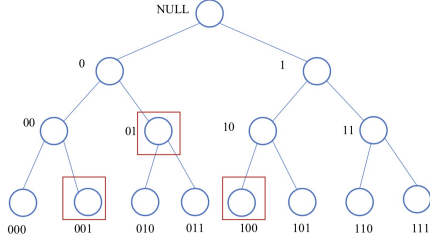


Fig. 1: The Example of the Canonical Covering of the Range $[001, 100]$ in T_3

A method $GetRoots(l, [a, b], \mathcal{K}_{[a, b]})$ is utilized to find the union of the canonical coverings of a set of ranges. The algorithm takes a tree T_l with $l + 1$ levels, a range $[a, b]$, and a subset of keys denoted as $\mathcal{K}_{[a, b]} = \{k_1, k_2, \dots, k_m\}$ where $\forall i \in [1, m], k_i \in [a, b]$. Assume the keys are in the ascending order. $GetRoots(l, [a, b], \mathcal{K}_{[a, b]})$ outputs a set of nodes R that represents the union of the canonical coverings of the ranges $[a, k_1 - 1], [k_1 + 1, k_2 - 1], \dots, [k_m + 1, b]$. Given $[a, b]$, those ranges are created by $\mathcal{K}_{[a, b]}$ and there is no key present in those ranges. Thus, we denote those ranges as empty ranges.

B. Hierarchical Identity-Based Encryption(HIBE)

We only care for the following property of HIBE which is important to our approach.

Given a hierarchy of identities arranged in a tree where identities correspond to the nodes, a secret key corresponding to a node z in the tree can be only used to decrypt any ciphertext intended for a node z' which is a node in the subtree rooted at z .

C. From HIBE to M-HIBE

A d -dimensional identity is denoted as $ID_1 || ID_2 || \dots || ID_d$ which is the concatenation of d identities. $\forall i \in [1, d]$, each ID_i corresponds to the identity in the i th dimension. In each dimension, there is a hierarchy of identities arranged in a tree where identities correspond to the nodes, as in HIBE. The property of the multi-dimensional hierarchical identity-based encryption scheme (M-HIBE) is that, a secret key corresponds to $ID_1 || ID_2 || \dots || ID_d$ can only be used to decrypt any ciphertext intended for $ID'_1 || ID'_2 || \dots || ID'_d$ in which for i ranging from 1 to d , ID'_i is a node in the subtree rooted at ID_i .

V. OUR PROPOSED APPROACH

Since our proposed approach is closely related to that of [1], we first review that construction as the preparation for our proposed approach.

A. Review on Ghosh et al.s work

Recall that the work in [1] addresses verifiable one-dimensional range queries over datasets of key-value pairs.

Let $\mathcal{D}$ denote a dataset of key-value pairs where each key-value pair is denoted as (k, v) . Assume the keys in $\mathcal{D}$ are $\mathcal{K} = \{k_1, \dots, k_n\}$ with the keys in the ascending order. Denote

$[a, b]$ as the queried range. Let $\mathcal{D}'$ denote the answer to the range query which contains key-value pairs whose keys are in $[a, b]$.

The proof of membership, i.e. $\mathcal{D}' \subseteq \mathcal{D}$, is straightforward so that we omit it. We concentrate on the protocol in [1] which is used for non-membership proof, that is, to prove that there is no $(k', v') \in \mathcal{D} \cap \mathcal{D}'$ such that $k' \in [a, b]$. More intuitively, the non-membership proof is to prove that the keys in $[a, b]$ which are not present in $\mathcal{D}'$ are not present in $\mathcal{D}$.

A trivial solution to the non-membership proof in zero-knowledge can be conducted as follows. For each key that is not present in $\mathcal{D}$, the data owner generates a secret key corresponding to the key using HIBE and sends it to the server. Upon receiving the answer $\mathcal{D}'$, the client generates the set of keys in $[a, b]$ which are not present in $\mathcal{D}'$. The client asks for the proof of knowledge of the secret keys corresponding to the set of keys he/she generates. The server proves the knowledge of those secret keys by decrypting the set of ciphertexts intended for the set of keys. Note that each ciphertext is intended for one key.

The set of keys that are not present in $\mathcal{D}$ can be represented by a set of ranges, so does the set of keys in $[a, b]$ which are not present in $\mathcal{D}'$. Recall that HIBE allows a secret key correspond to a node z in the tree to can be only used to decrypt any ciphertext intended for a node z' which is a node in the subtree rooted at z . Thus, for any range $[\alpha, \beta]$, the secret keys corresponding to the nodes in the canonical covering of the range can be used to decrypt any ciphertext intended for a key (i.e. a leaf node) in that range. For any range $[\alpha, \beta]$, the proof of knowledge of secret keys corresponding to the keys in $[\alpha, \beta]$ can be replaced by the proof of knowledge of secret keys corresponding to the canonical coverings of $[\alpha, \beta]$. The proof of knowledge of secret keys corresponding to keys in a set of ranges can be replaced by the proof of knowledge of secret keys corresponding to the union of the canonical coverings of the set of ranges.

By $GetRoots(l, [0^l, 2^l - 1, b], \mathcal{K})$, the data owner can find a set of nodes R which represents the union of the canonical coverings of the corresponding empty ranges generated by $\mathcal{K}$. By $GetRoots(l, [a, b], \mathcal{K}_{[a, b]})$, the client can find a set of nodes R^* which represents the union of the canonical coverings of the corresponding empty ranges generated by $\mathcal{K}_{[a, b]}$ which denotes the keys present in the answer $\mathcal{D}'$. Fig. 2 gives an illustration of the empty ranges generated by $\mathcal{K}$ and $\mathcal{K}_{[a, b]}$. We let the data owner generate secret keys corresponding to nodes in R and let the client ask for proof of knowledge of secret keys corresponding to nodes in R^* . Since each empty range generated by $\mathcal{K}_{[a, b]}$ is a subset of one of the empty ranges generated by $\mathcal{K}$, the data owner can prove the knowledge of secret keys corresponding to the nodes in R^* .

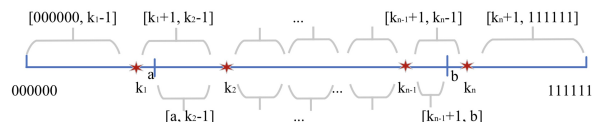


Fig. 2: An Illustration of Empty Ranges

B. Our proposed approach

As the first step, we construct a protocol for verifiable two-dimensional range queries in zero-knowledge using the similar technique in [1] while our construction is based on M-HIBE.

We assume the dataset $\mathcal{D}$ is a table containing n tuples with two attributes. Every tuple is formulated as a data point (x, y) in the 2D (two-dimensional) plane, where x, y are the values of the first and second attribute of the tuple, respectively. Assume each value is an l -bit binary string from 0^l to $2^l - 1$, similar to [1]. A 2D queried range is expressed as $[a, b] \times [c, d]$ which is a rectangle covering all points whose first coordinate is in $[a, b]$ and the second coordinate is in $[c, d]$. A correct answer should include all tuples (i.e. data points) in $\mathcal{D}$ in the rectangle $[a, b] \times [c, d]$. We call the 2D queried range $[a, b] \times [c, d]$ as a 2D (queried) range or a region or as a rectangle interchangeably, based on the context. Note that we also view a point $[a, b] \times [c, d]$ where $a = b, c = d$ or a line $[a, b] \times [c, d]$ where $a = b, c \neq d$ (or $a \neq b, c = d$) as rectangles (regions/2D ranges) for simplicity of description. Fig. 3 gives an example of the queried range $[011001, 110111] \times [001010, 101000]$ where $l = 6$.

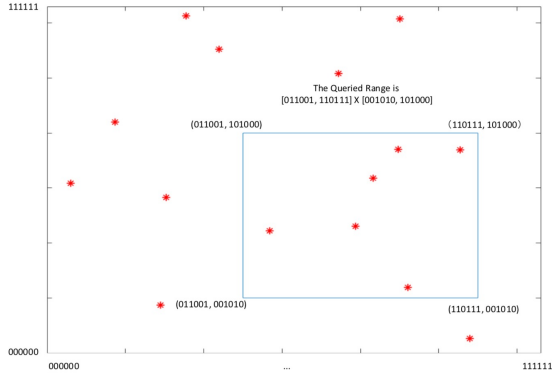


Fig. 3: An Illustration of the Queried Range

Given the 2D queried range $[a, b] \times [c, d]$, we need to prove the non-membership of data points in $[a, b] \times [c, d]$ excluding the data points in the answer.

Assume the answer to the query is $\mathcal{D}' = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$ with the first coordinate in the ascending order. We assume each x_i and y_i are distinct for simplicity of description where $i \in [1, m]$.

Like [1], we divide the set of data points in $[a, b] \times [c, d]$ that are not present in the answer into a set ER^* of disjoint empty regions, that is, $[a, x_1 - 1] \times [c, d]$, $[x_1, x_1] \times [c, y_1 - 1]$, $[x_1, x_1] \times [y_1 + 1, d]$, $[x_1 + 1, x_2 - 1] \times [c, d]$, $\dots$, $[x_m + 1, b] \times [c, d]$. Fig. 4 gives an example of those regions (the pink rectangles and the green lines), where the queried range is $[011001, 110111] \times [001010, 101000]$. To explain, those regions are generated due to the following facts. (1) The first coordinates presenting in the answer, i.e. $\{x_1, x_2, \dots, x_m\}$, show that the rectangles $[a, x_1 - 1] \times [c, d]$, $[x_1 + 1, x_2 - 1] \times [c, d]$, $\dots$, $[x_{m-1} + 1, x_m - 1] \times [c, d]$, $[x_m + 1, b] \times [c, d]$ are empty. (2) Based on the first coordinates and second coordinates presenting in the answer, i.e. $\{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$,

for every $x_i, i \in [1, m]$, the answer shows that the lines $[x_i, x_i] \times [c, y_i - 1]$ and $[x_i, x_i] \times [y_i + 1, d]$ are empty. Note that we only care for integers represented by l -bit binary strings in the 2D plane, thus there are some white spaces in Fig. 4 since the coordinates of data points in white spaces are not integers.

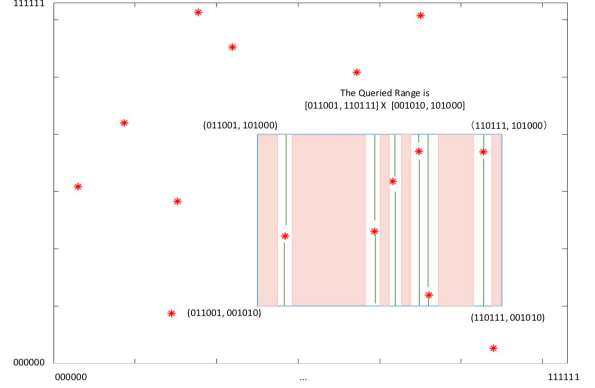


Fig. 4: An Illustration of the Set of Empty Regions

Based on the answer $\mathcal{D}'$, the client need the server to prove that ER^* (i.e. the set of empty regions) does not have data points in $\mathcal{D}$. Like the adoption of canonical covering of a range for proofs in [1], we utilize the canonical covering of a 2D range $[a, b] \times [c, d]$ which is defined as follows.

We first extend the node in the tree $\mathcal{T}_l$ (defined in Section IV-A) to a 2D node and define it. Then, we define the canonical covering of a 2D range based on 2D nodes.

Definition 1. The 2D node $z||w$: given two nodes z and w of $\mathcal{T}_l$, the 2D node $z||w$ is a presentation of the rectangle $[z(left), z(right)] \times [w(left), w(right)]$.

Definition 2. The canonical covering of a 2D range $[a, b] \times [c, d]$:

Assume the canonical covering of the range $[a, b]$ is $\{u_1, u_2, \dots, u_s\}$ where u_i is a node in $\mathcal{T}_l$, $\forall i \in [1, s]$. Assume the canonical covering of the range $[c, d]$ is $\{v_1, v_2, \dots, v_t\}$ where v_j is a node in $\mathcal{T}_l$, $\forall j \in [1, t]$. Then, the canonical covering of the 2D range $[a, b] \times [c, d]$ is the set S of 2D nodes denoted as $S = \{(u_i||v_j) \mid \forall i \in [1, s], j \in [1, t]\}$. $u_i||v_j$ is the 2D node which corresponds to the corresponding rectangle $[u_i(left), u_i(right)] \times [v_j(left), v_j(right)]$.

The rectangles represented by the 2D nodes are the full partition of the rectangle $[a, b] \times [c, d]$. For example, for the 2D range $[001, 100] \times [010, 110]$, the canonical covering of $[001, 100]$ is $\{001, 01, 100\}$ and the canonical covering of $[010, 110]$ is $\{01, 10, 110\}$. Therefore, the canonical covering of $[001, 100] \times [010, 110]$ is $\{001||01, 001||10, 001||110, 01||01, 01||10, 01||110, 100||01, 100||10, 100||110\}$, as shown by the green colored regions in Fig. 5. The white spaces do not contain any valid data points whose coordinates are integers, thus we do not care for it.

Note that every 2D node $u_i||v_j$ can be viewed as a 2D identity in M-HIBE, where $i \in [1, s], j \in [1, t]$. The secret key for this 2D node can be used to decrypt the ciphertext intended

for any $u_i || v_j$ where u_i is a node in the subtree rooted at u_i and at the same time v_i is a node in the subtree rooted at v_i .

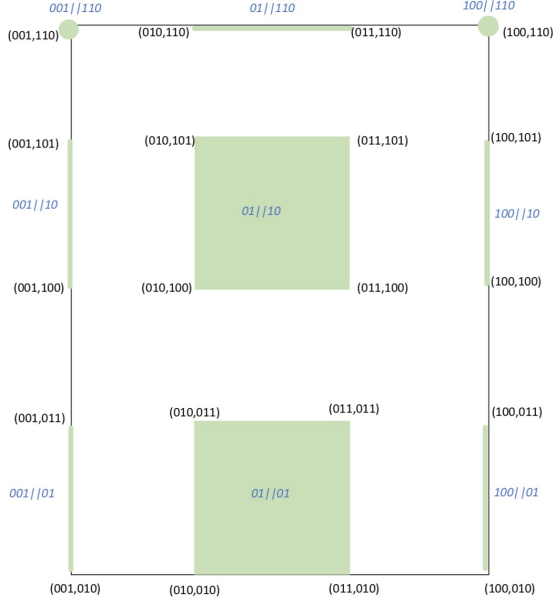


Fig. 5: An Illustration of the Canonical Covering of a 2D Range $[001, 100] \times [010, 110]$

Recall that ER^* is the set of empty regions. Assume the data points in $\mathcal{D}$ is $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ with the first coordinate in the ascending order. Like [1], we let data owners generate ER with by the same principle of generating ER^* , with the given range $[0^l, 2^l - 1] \times [0^l, 2^l - 1]$. Let Q^* and Q be the union of the canonical coverings of the empty regions in ER^* and ER respectively. We expect to transform the proof of non-membership of data points in regions in ER^* to the proof of the knowledge of M-HIBE secret keys corresponding to Q^* . That is, iff the server can prove the knowledge of secret keys corresponding to Q^* , the client is convinced that the regions in ER^* do not have points in $\mathcal{D}$.

Unfortunately, unlike [1] where each empty range generated by $\mathcal{K}$ is a subset of one of the empty ranges generated by $\mathcal{K}_{[a,b]}$, an empty region in ER^* generated by $\mathcal{D}'$ in the 2D environment is not always a subregion of one of the empty regions in ER generated by $\mathcal{D}$. For example, assume $\mathcal{D} = \{(x_1, y_1), (x_2, y_2), \dots, (x_{n-1}, y_{n-1})\}$, $x_n \in [a, b]$ and $y_n < c$. With regard to x_{n-1} and x_n , the data owner will generate secret keys for the canonical covering of the regions $[x_{n-1}, x_{n-1}] \times [0^l, y_{n-1} - 1]$, $[x_{n-1}, x_{n-1}] \times [y_{n-1} - 1, 2^l - 1]$, $[x_{n-1} + 1, x_n - 1] \times [0^l, 2^l - 1]$, $[x_n, x_n] \times [0^l, y_n - 1]$, $[x_n, x_n] \times [y_n + 1, 2^l - 1]$, and $[x_n + 1, 2^l - 1] \times [0^l, 2^l - 1]$. However, since (x_{n-1}, y_{n-1}) is the last data point in the answer, the client asks for proof of knowledge of secret keys corresponding to the canonical covering of $[x_{n-1} + 1, b] \times [c, d]$. Assume the canonical covering of $[x_{n-1} + 1, b]$ is $\{u_1, \dots, u_s\}$ and the canonical covering of $[c, d]$ is $\{v_1, \dots, v_t\}$. Since $x_n \in [x_{n-1} + 1, b]$, x_n is a node in the subtree rooted at one of the nodes. e.g. u_k where $k \in [1, s]$. Since $x_n \notin [x_{n-1} + 1, x_n - 1]$ and $x_n \notin [x_n + 1, b]$, the u_k is not a node in

the subtree rooted at any of the nodes in the canonical covering of $[x_{n-1} + 1, x_n - 1]$ or $[x_n + 1, b]$. If $x_n \neq u_k$ which is usually true, the server will not have the knowledge of secret keys for 2D nodes $\{u_k || v_j\}_{j \in [1, t]}$. Thus, the server cannot convince the client of the non-membership of data points in $[x_{n-1} + 1, b] \times [c, d]$.

To make the server have the of knowledge of secret keys corresponding to the canonical coverings of the region $[x_{n-1}, x_n] \times [c, d]$, our solution is let the data owner leak more secret keys to the sever. As long as the data owner only leak secret keys for the canonical coverings of regions that do not have data points in $\mathcal{D}$, the server cannot manage to let the client believe that a region that has data points does not have data points present in $\mathcal{D}$.

We observe that for u_i , $i \in [1, s]$, u_i satisfies either of the following conditions:

- 1) there is no data point in $\mathcal{D}$ whose first coordinate is in the range $[u_i(left), u_i(right)]$.
- 2) there is one or more data points in $\mathcal{D}$ whose first coordinates are in the range $[u_i(left), u_i(right)]$.

For any of the cases, the client asks for the proof of knowledge of secret keys corresponding to the canonical covering of the region $[u_i(left), u_i(right)] \times [c, d]$. If the first condition holds for all u_i where $i \in [1, s]$, i.e. there is no (x_n, y_n) in $\mathcal{D}$, the 2D range $[x_{n-1} + 1, b] \times [0^l, 2^l - 1]$ should be in ER and thus the secret keys for $[x_{n-1} + 1, b] \times [0^l, 2^l - 1]$ can be used to prove the knowledge of secret keys for $[u_i(left), u_i(right)] \times [c, d]$, since $[u_i(left), u_i(right)] \times [c, d]$ is a subregion of $[x_{n-1} + 1, b] \times [0^l, 2^l - 1]$.

If the second condition holds for some u_i where $i \in [1, s]$, at least there is some $x_n \in [u_i(left), u_i(right)]$. We want the server to have the knowledge of secret keys corresponding to the canonical covering of the region $[u_i(left), u_i(right)] \times [c, d]$. Since $[u_i(left), u_i(right)] \times [c, d]$ is not a subregion in any region in ER , we need to let the data owner find an empty region of which $[u_i(left), u_i(right)] \times [c, d]$ is a subregion and then generate secret keys corresponding to the canonical covering of that region. We define $PR(u_i)$ as a set of points in $\mathcal{D}$ whose first coordinates are within $[u_i(left), u_i(right)]$. Assume $PR(u_i) = \{(x_1, y_1), \dots, (x_m, y_m)\}$ where the first coordinates are in the ascending order. Given the 2D range $[u_i(left), u_i(right)] \times [0^l, 2^l - 1]$, based on $PR(u_i)$, we generate a set ER' of regions as $[u_i(left), u_i(right)] \times [0^l, y_1 - 1]$, $[u_i(left), u_i(right)] \times [y_1 + 1, y_2 - 1]$, $\dots$, $[u_i(left), u_i(right)] \times [y_m + 1, 2^l - 1]$. ER' is generated by the similar principle of ER and ER^* , with the difference that we generate empty rectangles by the second coordinates in $PR(u_i)$. If $[u_i(left), u_i(right)] \times [c, d]$ is empty, there must be a region of which $[u_i(left), u_i(right)] \times [c, d]$ is a subregion.

We observe that each u_i for which the second condition holds is the node that some x_j is a descendant of or is equal to, where $(x_j, y_j) \in \mathcal{D}$ and $j \in [1, n]$. Therefore, $\forall j \in [1, n]$, we can do the above procedure for all the nodes which are ancestors of the first coordinates present in $\mathcal{D}$.

For each empty region in ER , the server needs to store $O(l^2)$ secret keys. In ER , there are $O(n)$ regions. Thus, the number of secret keys for ER is $O(nl^2)$.

For each empty region in ER' , the server needs to store $O(l)$ secret keys. For each u_i which is an ancestor of a first coordinate x_j present in $\mathcal{D}$, the number of regions in ER' is $O(|PR(u_i)|)$. In total, the number of regions in ER' for all u_i s which are ancestors of the first coordinates present in $\mathcal{D}$ is $O(nl)$. Thus, the number of all empty regions is $O(nl)$ and the server need to store $O(nl^2)$ secret keys.

The total number of secret keys to be stored is $O(nl^2)$.

VI. PROPOSED WORK

Our proposed work is as follows.

- 1) Develop an efficient construction of an M-HIBE scheme. The approach we have outlined in the previous section assumes the existence of a secure M-HIBE scheme.
- 2) Prove that our approach satisfies the security requirements.
- 3) Analyze in detail the time and space complexity of the algorithms executed by each party and the space used by the proofs.
- 4) Extend our construction to d -dimensional range queries for $d > 2$.
- 5) Perform experiments to test the practical efficiency of our protocol using real-world datasets.

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